D1. Literature and Technology Review

Technology review for AI-based cybersecurity audit assessment against a benchmark.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

What was reviewed and why

This deliverable is a technology review, not an experiment report. It asks which RAG capabilities matter for evidence-bound cybersecurity assessment: provenance-preserving parsing, hybrid retrieval, controlled generation, and measurable evaluation. The review separates published approaches from repository implementation and from completed measurements so a reader can see both ambition and evidence. [E8][S1][S2]

The recommended architecture is a practical layered design rather than a claim that every new RAG component should be switched on. Text/JSON evidence comes first; visual retrieval is justified only for page-only evidence; graph retrieval is justified only for relationship or cross-report questions. This avoids expensive components where a local factual lookup is sufficient. [S2][S3][S6]

Evidence maturity of reviewed technologies

Technology	Published rationale	Project status
Hybrid dense + BM25 retrieval	Semantic and exact-term retrieval complement each other	Implemented and ablated [E10]
Modern Qwen embedding	Local multilingual retrieval candidate	Used in completed runs [E4][E10]
Cross-encoder reranking	Improves final relevance ordering	Implemented; not enabled in 6 GB production run [E8]
Late interaction / visual RAG	Useful for technical phrase or page-layout evidence	Integration path only; not a measured claim [E8][S2][S3]
Graph-assisted retrieval	Useful for entity and cross-report relations	Optional route; no isolated effectiveness claim [E4][S6]

Scope and conclusion

This review distinguishes published architecture options from work that was actually implemented and tested in this repository. The implemented baseline is a text-first hybrid RAG pipeline; it is not presented as a proven state-of-the-art system. [E8]

The practical recommendation is structured hybrid retrieval as the default. Visual retrieval should be enabled only when page images, tables or screenshots are in the accepted input scope; graph retrieval should be reserved for cross-report relationship questions. [S2][S3][S6]

Technology comparison

Family	Contribution	Project use	Evidence status
Dense plus sparse hybrid	Semantic retrieval plus exact technical terms	Baseline B0: embeddings, BM25 and rank fusion	Implemented; saved runs exist [E1][E4]

Family	Contribution	Project use	Evidence status
Cross-encoder reranking	Second-stage relevance ordering	Implemented; disabled in the two-report GPU run	Code inspected; no two-report reranker result [E4][E8]
Qwen3 embedding/reranking	Modern local retrieval candidates	Downloaded low-resource models	Retrieval comparison completed [E2]
Late interaction	Token-level technical-term matching	Candidate for identifiers/OCR noise	Implemented integration point; not evaluated [E8][S2]
Visual document retrieval	Page/table/screenshot retrieval	Out of current text-only assessment scope	CLI path only; not evaluated [E8][S3]
Graph-assisted retrieval	Finding/entity relationship context	Optional cross-report route	Enabled in real-report configuration; no isolated ablation [E4][S6]

Selection implications

Published capability is not selection evidence. The final choice must be supported by identical corpus, labels, retrieval budget, generator settings, hardware and measured quality/safety metrics. [S1][S4][E8]

The present evidence supports using the repository as an auditable proof of concept. It does not support claims of superiority over alternative model families or operational performance on unseen audit agencies. [E4][E5]

Evidence and sources

Evidence citations used in this report are listed below. External sources describe candidate technologies; repository artifacts support implementation and measured-result statements.

Technology adoption decision

Capability	Decision for this POC	Reason and review condition
Hybrid retrieval	Adopt	Completed dense-only versus hybrid comparison improved answer accuracy from 50.00% to 62.50%. [E10]
Evidence-first prompting	Adopt	v2 improved the Qwen2.5 control and makes evidence IDs machine-checkable. [E10]
Reranking	Defer	Code path exists, but it was disabled for the 6 GB laptop run; no claimed benefit. [E4][E8]
Visual / graph retrieval	Constrain	Visual input is outside scope; graph route has no isolated effectiveness result. [E4][E8]

Review conclusion

The technical position is intentionally selective. The submission does not portray every available RAG component as an achievement; it adopts only components supported by a concrete need or a completed local comparison. This makes the architecture credible for an audit-review setting, where explainability and repeatability matter more than novelty. [E8][E10]

The resulting research conclusion is that a local, provenance-preserving hybrid RAG design is a defensible POC foundation. Reranking, visual retrieval and richer graph reasoning remain bounded future experiments with their own acceptance criteria, not implied production features.

Repository evidence cited

- [E1] src/results/1787426063_6d47ba9f8ee0/: saved baseline run artifacts (one report, 12 questions).
- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.

- [E8] src/{docling_parser.py,chunking.py,embed_index.py,pipeline.py,evaluation_harness.py,adaptive_rag.py,prompts.py,icl.py}: implementation inspection.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.

External technology sources

- [S1] Chen et al. (2024), BGE-M3-Embedding, arXiv:2402.03216, https://arxiv.org/abs/2402.03216.
- [S2] Santhanam et al. (2021), ColBERTv2, arXiv:2112.01488, https://arxiv.org/abs/2112.01488.
- [S3] Faysse et al. (2025), ColPali, arXiv:2407.01449, https://arxiv.org/abs/2407.01449.
- [S4] Yang et al. (2025), Qwen3 Technical Report, arXiv:2505.09388, https://arxiv.org/abs/2505.09388.
- [S5] Qwen (2025), Qwen3-Reranker-4B model card, https://huggingface.co/Qwen/Qwen3-Reranker-4B.
- [S6] Microsoft GraphRAG query documentation, https://microsoft.github.io/graphrag/query/overview/.